Modelling Stock Market Volatility: a GARCH approach for the DAX Index

Econometrics Final Project

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Group 01

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1. Introduction

In today's complex and dynamic financial markets, predicting stock market volatility has become crucial for investors, financial institutions, and policymakers. Volatility forecasting provides valuable insights into the future behaviour of financial markets, enabling stakeholders to make informed decisions and manage risks effectively. This project focuses on forecasting stock market volatility using the DAX index, a key benchmark for the German equity market.

This project aims to employ various econometric models and compare their performance forecasting stock market volatility. Specifically, we utilized four well-known models: Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model, GARCH in Mean (GARCH-M), Exponential GARCH (E-GARCH), and Glosten-Jagannathan-Runkle GARCH (GJR-GARCH). These models are widely recognized and extensively used in economic research due to their ability to capture important characteristics of financial time series.

By utilizing historical data of the DAX index, we constructed and estimated each model to generate volatility forecasts for December 2022. Forecasting can be useful to investors seeking to optimize their portfolio allocation, hedge against market risks, or implement trading strategies based on anticipated volatility levels.

The comparison of these models offers valuable insights into their relative performance in terms of accuracy and parsimony in capturing the intricate nature of stock market volatility. Each model has its unique features and assumptions, and understanding its strengths and weaknesses can help market participants make more informed decisions based on their specific needs and risk preferences.

The following section explores the literature review (see Section 2), showcasing the field's current state. Additionally, we discuss the dataset employed in this study (see Section 3) and outline the methodology utilized (see Section 4). Furthermore, we present the results and volatility forecasts (see Section 5), culminating in a summary of the key findings of this research (see Section 6).

2. Literature Review

As a leading benchmark for the German stock market, the DAX Index exhibits significant volatility that poses challenges and opportunities for investors and researchers alike. To effectively analyse and predict an equity index's volatility, we exploit sophisticated models that can capture the so-called stylised facts of volatility in our work. These stylised facts are well documented in the literature. The following features are common to many univariate financial time series:

- *Fat tails*: the distribution of financial returns exhibits fatter tails than the normal distribution;
- *Volatility clustering*: financial volatility tends to cluster, meaning that a large (small) price change tends to be followed by another large (small) price change;
- *Leverage effect*: an unexpected price drop increases volatility more than an unexpected price increase of equal magnitude;
- *Long memory effect*: financial market volatility is characterised by long-range dependencies.

This literature review aims to provide an overview of existing research that explores the application of these models in understanding and forecasting the volatility of a stock index.

Engle in [7] and Bollerslev in [5] developed a new class of volatility models, the Generalised Autoregressive Conditional Heteroskedasticity (GARCH) models, that can capture the volatility clustering effect. In the basic GARCH model, the conditional variance depends only on its lags and lags of squared innovations. The concept was then extended by Nelson in [15], Glosten et al. in [9], and Zakoian in [19], among others, to model additional empirical characteristics of volatility.

In the literature, the comparative analysis of GARCH-type models in volatility forecasting has provided valuable insights into their relative performance and suitability for different financial markets.

For example, Engle et al. in [8] studied the Japanese stock market by simultaneously applying the E-GARCH, AGARCH, NGARCH, VGARCH and GJR-GARCH models in the research. They found by empirical analysis that the

Japanese stock market had volatility asymmetry, and the conditional variance estimated by the E-GARCH model was generally more significant than other models. Still, the GJR-GARCH model did the best in reflecting the asymmetric effect. In contrast, Yeh and Lee in [18] doubted this result: they tested the GJR-GARCH model on the Chinese stock market and found that this model generates a higher estimated conditional variance during the period of high volatility.

Another example of comparative analysis of GARCH-type models regards the volatility of the S&P 500 index. Najand in [14] estimated the S&P 500 returns data and found that in stock volatility forecast, sometimes asymmetric models like E-GARCH, TRARCH etc. provide more fitness than GARCH(1,1), GARCH-M such symmetric models. By using different GARCH models to analyse the volatility of the S&P 500 Index, Awartani and Corradi in [3] found that under the premise of symmetric information, GARCH (1,1) can accurately describe the volatility of the data. On the other hand, the asymmetric GARCH models would be more suitable under the circumstances of asymmetric information.

While no single GARCH model emerges as universally superior, researchers have identified certain models that excel in capturing specific characteristics, such as leverage effects or asymmetry. Selecting the appropriate GARCH model depends on the particular objectives, the features of the data, and the dynamics of the financial market under consideration. Further research and advancements in model selection techniques and hybrid approaches will continue to enhance volatility forecasting accuracy in the future.

3. Data

3.1. Dataset desciption

The data used for our study are the daily adjusted closing prices of the DAX Index (i.e. prices are adjusted for splits and dividend and/or capital gain distributions). The dataset provides information over the period extending from the 3^{rd} of January 2008 to the 31^{st} December 2022, i.e. 3804 observations, one for each trading day. The DAX (Deutscher Aktien Index) is composed of the largest capitalized and most liquid German stocks listed on the *Prime Standard* segment of the Frankfurt Stock. Therefore, we have opted to examine this index being an indicator of the German economy's performance, its provision of investment opportunities, its influence on global markets, and its role as a benchmark for diverse financial products.
In Figure 1, the historical series of DAX price is represented in the observation period.

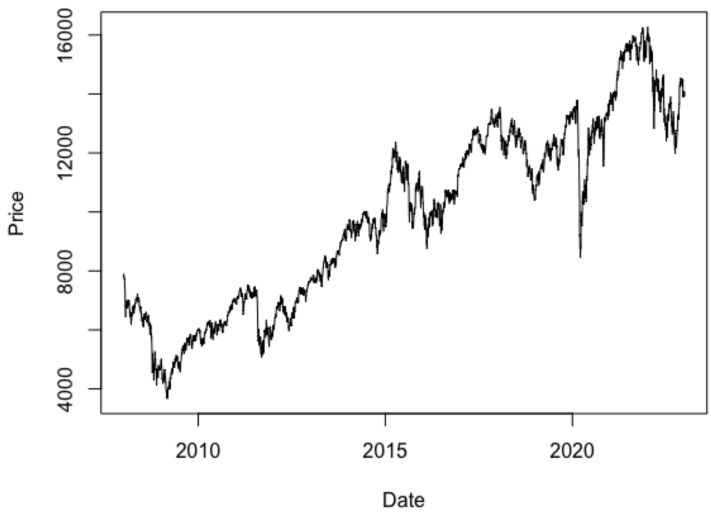


Figure 1: DAX daily price in €

The volatility analysis is done not for the DAX, but for historical series of log-return with daily observation frequency, defined as:

$$r_t = log\left(\frac{S_t}{S_{t-1}}\right), \tag{1}$$

where S_t is the closing value of DAX at day t.

3.2. Empirical Analysis

In this section, we provide a brief overview of the key stages involved in the initial data analysis, aiming to acquire a deeper understanding of the modelled index. Throughout this analysis, we will focus on exploring several critical aspects of the financial time series, including descriptive statistics, stationarity, autocorrelation, and heteroskedasticity. By examining these elements, we aim to uncover valuable insights that will contribute to a more robust analysis of the data.

3.2.1. Descriptive statistics

As a preliminary step, it is necessary to have an overview of the basic statistical features of the log-return time series. Firstly, the trend of the log-returns over the entire period considered is analyzed and represented in Figure 2.

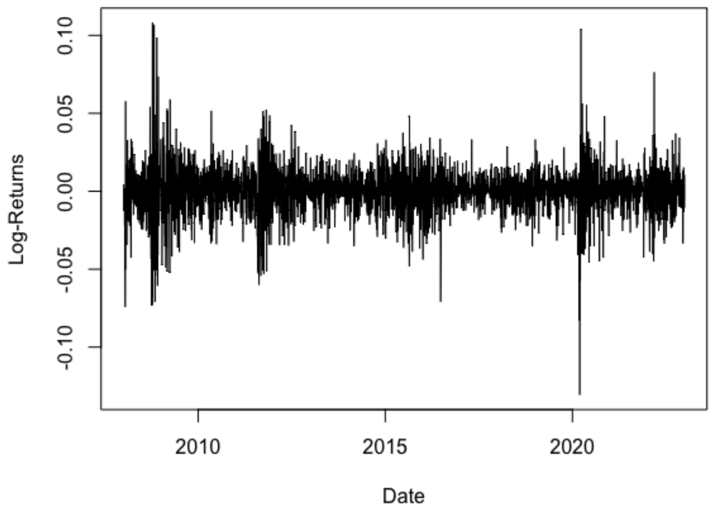


Figure 2: Graph of daily DAX index log-return

The chart shows apparent features of time-varying variance and the presence of clusters that persist over time (volatility clustering), where periods of low volatility are followed by periods of high volatility. Hence, the traditional conditional variance models with the assumption of homoskedasticity may not be the most suitable for fitting the volatility of the DAX Index. Instead, ARCH and GARCH models may properly do this job, since they are capable of dealing with time series in the presence of heteroskedasticity and clustering.

Table 1 reports the standard descriptive statistics of the time series of the daily log-return of the DAX.

Samples	Mean	Median	Max	Min	Std.Dev.	Skewness	Kurtosis
3804	0.000158	0.000726	0.107975	-0.130549	0.014250	-0.189804	10.7051

Table 1: Descriptive statistics for DAX log-return

From the statistics, it can be observed that the mean of the return series is approximately 0. This is expected since return series usually have a regressive tendency towards a long-term value. Moreover, the gap between the maximum and minimum values is 0.2385, and the standard deviation is around 1.07%: these two values imply relatively high volatility for the index during our sample period. The negative value of skewness may be due to the negative inclination of the asymmetric tail, instead, the kurtosis value (10.7051) is much greater than the one standard normal distribution value (3): these values reveal that the distribution of DAX has characteristics of "fat tail" and "sharp peak" (*leptokurtic*).

Now, the focus is on analyzing the distribution type followed by the sequence of log-returns. The graph presented below displays the empirical distribution of the time series, together with a Normal distribution characterized by the mean and standard deviation of the analyzed data.

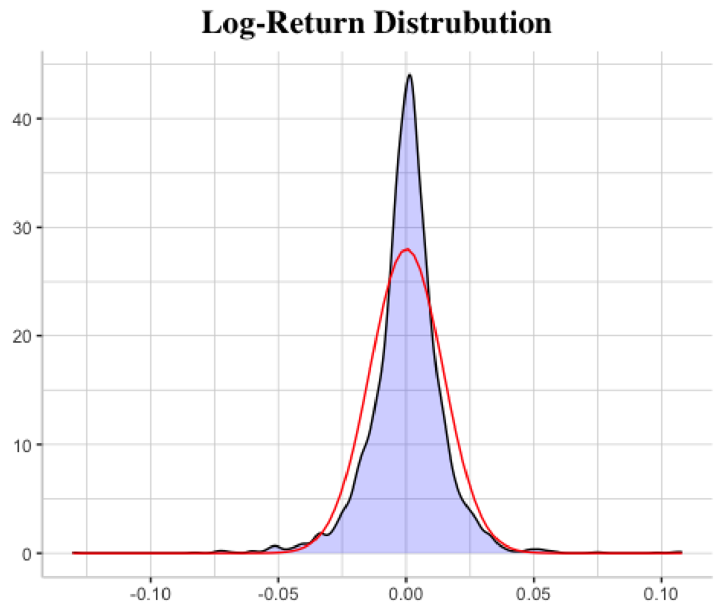


Figure 3: Distribution of the daily log-return of the DAX index vs Normal Distribution

Based on the obtained results, it has been decided to proceed with several statistical tests to assess the Gaussianity of the data. In particular, the following tests have been conducted:

- *Shapiro-Wilk* test [16];
- *Anderson-Darling* test [17];
- *Kolmogorov-Smirnov* test [13];
- *Jarque-Bera* test [10].

The p-values obtained from all of these tests are reported and found to be less than 0.05. This indicates that the null hypothesis can be rejected, suggesting that the return series is not subject to a Normal distribution.

	p-value
Shapiro-Wilk test	3.462782e-40
Anderson-Darling test	3.7e-24
Kolmogorov-Smirnov test	0
Jarque-Bera test	0

Table 2: Gaussianity test

3.2.2. Stationarity

Next, the stationarity of the return series is examined by considering the Augmented Dickey–Fuller (ADF) test [6]. The test results, as shown in Table 3, indicate that the hypothesis of a Unit Root can be rejected at the 5% level of significance.

	p-value
Augmented Dickey–Fuller Test	0.01

Table 3: Stationarity test

3.2.3. Heteroskedasticity

Accepting the hypothesis of homoskedasticity introduces a strongly distorting element in the estimation of model parameters and related tests in the analysis of the series. Empirical analyses, in fact, demonstrate that a significant portion of financial series exhibits heteroskedasticity: a historical series is considered heteroskedastic if it exhibits a variance characterized by a non-constant behaviour.

In this section, the analyses conducted are presented. The decision was made to conduct the *Bartlett's test* [4], which is a statistical test used to assess whether there are statistically significant differences in the variances of the residuals across various groups or conditions. A high p-value obtained from the Bartlett test suggests that there is insufficient evidence to reject the hypothesis of homoskedasticity. Conversely, a low p-value provides statistical evidence to reject the hypothesis of homoskedasticity, indicating substantial variations in the variances among the examined groups.

However, prior to that, it is fundamental to identify the structural break point, representing a significant change in our data. The approach used is the Pruned Exact Linear Time ($PELT$) which is an algorithm computationally efficient that determines the optimal locations of breaks in the data. It prunes the search space by discarding candidate breakpoints that do not significantly improve the overall fit of the model. This method allows to detect and locate significant changes in the underlying patterns, trends, or parameters of the data series.

In Figure 4 can be observed 12 structural breaks points were identified, in particular, four distinct periods are highlighted within the considered time frame:

- The first refers to the financial crisis of 2008 and the subsequent attempts at economic recovery by the German government;
- The second indicates the sovereign debt crisis of the Eurozone in the years 2011-2012;
- The third 2014-2016 one pertains to the occurrence of Brexit;
- The fourth is the period during which global financial markets underwent a phase of significant volatility and turbulence due to the COVID-19 emergency.

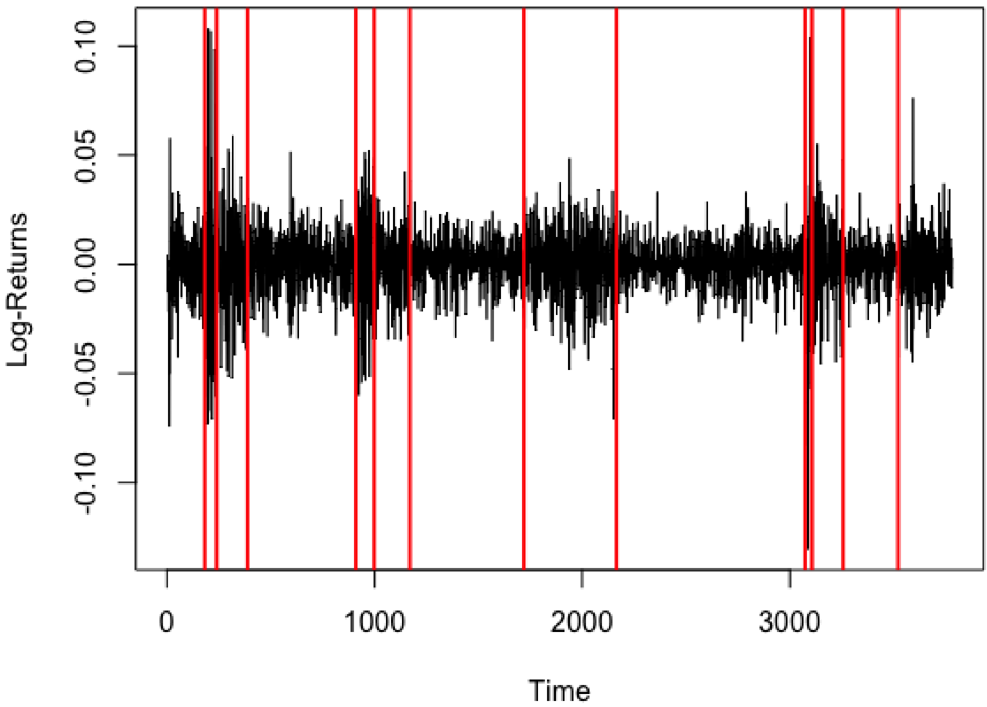


Figure 4: Breaks point detection using PELT method

At this point, Bartlett's test is applied to the time series divided into groups identified by the structural break points with the aim of comparing the variances of the residuals among this subset.
Below, the result of the test is reported:

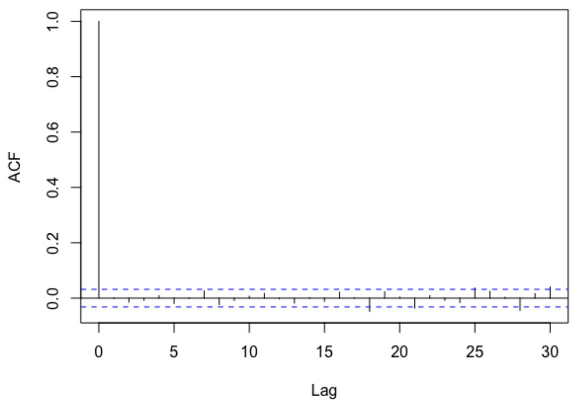
	p-value
Bartlett's Test	2.2e-16

Table 4: Heteroskedasticity test

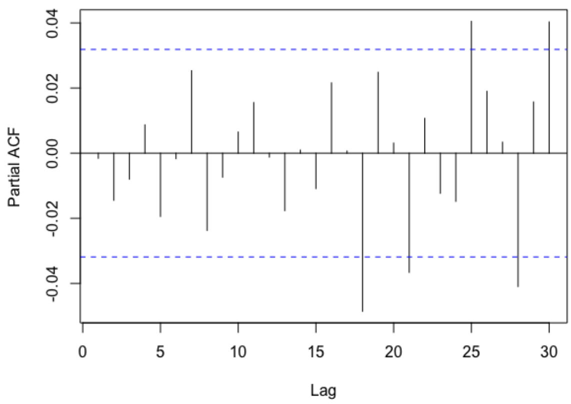
The p-value is less than the chosen significance level of 5%, so it can be concluded that the variances between the groups are significantly different, indicating heteroskedasticity in the data.

3.2.4. Autocorrelation and ARCH Effect

The economic and financial series are used to identify recurring patterns that can be used to predict price fluctuations. This corresponds to the fundamental concept of autocorrelation, which measures the relationship between the past values of a time series and its future values. Autocorrelation can be assessed using various methods as the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots, which are presented below:



(a) Autocorrelation Function (ACF).

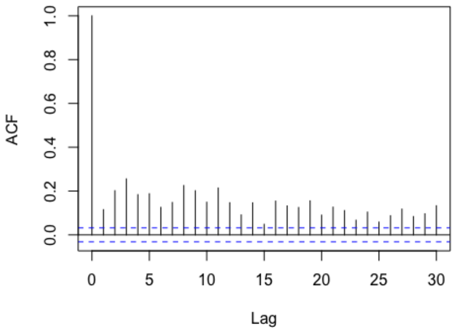


(b) Partial Autocorrelation Function (PACF).

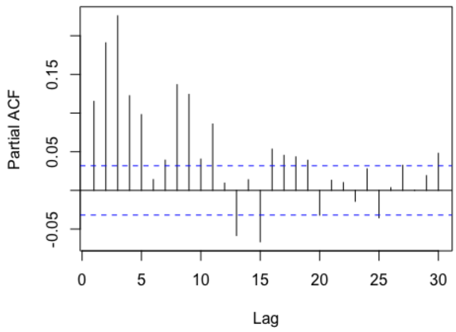
a. The Autocorrelation Function plot demonstrates how the autocorrelation value of simple returns falls within the 95% confidence bands after the first lag, thus indicating the absence of autocorrelation in the returns so that no Moving Average behaviour can be detected. Additionally, the ACF graph displays values with opposite signs, this suggests a mean-reverting process, where new information only affects price changes in the short term.

b. The Partial Autocorrelation Function plot suggests that no autoregressive behaviour is present in the time series because the only significant lags are meaningless from a financial perspective. Thus, combining the information from the ACF and PACF plots, we can conclude that the mean structure of the log returns does not show an ARMA behaviour.

The absence of autocorrelation in simple log-returns is not maintained when we analyzed squared residual approximation for volatility. The following graphs show this:



(c) Autocorrelation Function of squared residuals (ACF).



(d) Partial Autocorrelation Function of squared residuals (PACF).

The autocorrelation function (ACF) and partial autocorrelation function (PACF) for the series of squared daily log-returns of the DAX series reveal the presence of *ARCH effects*.
This phenomenon has been validated by the Arch test, of which the results are subsequently reported:

Lag	**p-value**
1	1.156e-12
2	$<$ 2.2e-16
3	$<$ 2.2e-16
4	$<$ 2.2e-16

Table 5: ARCH Effect test

The p-value is less than 0.05, so the null hypothesis is rejected, indicating the presence of a significant ARCH effect. This confirms the existence of conditional heteroskedasticity in the time series.

Moreover, both graphs demonstrate an extended tail, suggesting that the ARCH(p) model may not be suitable. A higher value of p would be required to describe the phenomenon accurately. Hence, the GARCH(1,1) model estimation is preferred so that it will be the starting point for our analysis.

4. Methodology

The purpose of our work is to select the most accurate and parsimonious model to capture the volatility structure of our daily price data. In order to archive this result, the main GARCH models have been evaluated using different orders of dependence upon previous variance lags and squared error terms. The best order structure for every GARCH model has been selected according to some Information Criteria such as Akaike and Bayes. Finally, their prediction power has been tested forecasting the volatility in the month of December 2022 and comparing it with the actual realizations of the log returns. The fitted models are the standard GARCH, the E-GARCH, the GJR-GARCH and the GARCH-M, and they have been evaluated using one, two and three lags for the squared error terms and only one for the autoregressive part. Each of them tries to capture different features of the behaviour of the volatility of the returns, such as asymmetry, leverage effect, leptokurtosis, volatility clustering and the presence of risk premium in the presence of high volatility.
For the first three models, the mean structure is the following:

$$y_t = \mu + u_t \qquad u_t \sim N(0, \sigma_t^2) \tag{2}$$

and the variance structures are:

GARCH(1,q)

$$\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i u_{t-i}^2 + \beta \sigma_{t-1}^2 \qquad q = 1, 2, 3; \tag{3}$$

E-GARCH(1,q)

$$log(\sigma_t^2) = \omega + \beta log(\sigma_{t-1}^2) + \sum_{i=1}^{q} \left(\gamma_i \frac{u_{t-i}}{\sqrt{\sigma_{t-i}^2}} + \alpha_i \frac{|u_{t-i}|}{\sqrt{\sigma_{t-i}^2}} \right) \qquad q = 1, 2, 3; \tag{4}$$

GJR-GARCH(1,q)

$$\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i u_{t-i}^2 + \beta \sigma_{t-1}^2 + \gamma u_{t-1}^2 I_{t-1} \qquad q = 1, 2, 3; \tag{5}$$

$$I_{t-1} = I\left\{u_{t-1} < 0\right\};$$

For the **GARCH-M(1,q)** model, the variance structure is the same as a standard GARCH model, but the mean design is different.

$$y_t = \mu + u_t + \delta\sigma_{t-1} \qquad u_t \sim N(0, \sigma_t^2) \tag{6}$$

$$\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i u_{t-i}^2 + \beta\sigma_{t-1}^2 \qquad q = 1, 2, 3; \tag{7}$$

After evaluating and choosing the best order for every model, it has been necessary to check the assumptions for the standardized residuals obtained as the ratio between the actual residuals and the estimated standard deviations. The hypotheses to be verified are Gaussianity and the absence of autocorrelation between the standardized residuals.

$$v_t = \frac{u_t}{\sigma_t} \sim N(0, 1) \tag{8}$$

$$E\left[v_t v_{t-\tau}\right] = 0 \quad \forall\tau \neq 0 \tag{9}$$

5. Results

In this section, results of the fitted GARCH models and models' coefficients (Subsection 5.1), results coming from the diagnostic (Subsection 5.2) and from the forecasting (Subsection 5.3) are reported.

5.1. Models and Coefficients

Model and order	AIC	BIC
GARCH(1,1)	-5.9884	-5.9818
GARCH(1,2)	-5.9893	-5.9810
GARCH(1,3)	-5.9890	-5.9791
E-GARCH(1,1)	-6.0399	-6.0316
E-GARCH(1,2)	-6.0444	-6.0328
E-GARCH(1,3)	-6.0448	-6.0299
GJR-GARCH(1,1)	-6.0313	-6.0230
GJR-GARCH(1,2)	-6.0304	-6.0188
GJR-GARCH(1,3)	-6.0299	-6.0150
GARCH-M(1,1)	-5.9891	-5.9809
GARCH-M(1,2)	-5.9899	-5.9800
GARCH-M(1,3)	-5.9896	-5.9781

Table 6: Values of model selection Criteria indices.

The suggested order for every model is different depending on the Criterion which is used. The weak differences in the numerical results and the need for an easy-to-interpret model have led to the choice towards the easiest model for every GARCH extension, that is to say, GARCH(1,1), E-GARCH(1,1), GJR-GARCH(1,1) and GARCH-M(1,1). The optimal parameters and their significativities are reported for every model.

GARCH(1,1)	μ	ω	α_1	β_1
Optimal Parameter	0.000651 *	0.000004 *	0.098961 *	0.882964 *
Robust Parameter	0.000651 *	0.000004	0.098961 *	0.882964 *

Table 7: Parameters of the GARCH(1,1) model. * denotes significance at the 5% level, the usual benchmark in the econometrics literature.

E-GARCH(1,1)	μ	ω	α_1	β_1	γ_1
Optimal Parameter	0.000098	-0.226281 *	-0.130570*	0.973932 *	0.123474 *
Robust Parameter	0.000098 *	-0.226281 *	-0.130570*	0.973932 *	0.123474 *

Table 8: Parameters of the E-GARCH(1,1) model. * denotes significance at the 5% level, the usual benchmark in the econometrics literature.

GJR-GARCH(1,1)	μ	ω	α_1	β_1	γ_1
Optimal Parameter	0.000221	0.000003 *	0.000000	0.901579 *	0.154991 *
Robust Parameter	0.000221	0.000003 *	0.000000	0.901579 *	0.154991 *

Table 9: Parameters of the GJR-GARCH(1,1) model. * denotes significance at the 5% level, the usual benchmark in the econometrics literature.

GARCH-M(1,1)	μ	δ	ω	α_1	β_1
Optimal Parameter	-0.000587	0.118484*	0.000004 *	0.098677 *	0.882731*
Robust Parameter	-0.000587	0.118484*	0.000004	0.098677 *	0.882731*

Table 10: Parameters of the GARCH-M(1,1) model. * denotes significance at the 5% level, the usual benchmark in the econometrics literature.

The non-negativity constraint is not violated for any model. The reason for the development of the standard GARCH model over the simpler ARCH model is the possibility of using a more parsimonious model less likely to violate this restriction. The only negative coefficients can be found in the E-GARCH model, but this fact does not break any constraint since the model is describing the logarithm of the volatility. The modelling of the logarithm of the variance is useful in order to avoid the non-negativity issue.

It is possible to notice that the estimated parameters for the volatility structures of the standard GARCH and of the GARCH-M are only slightly different, probably due to numerical approximations. This is what we would expect since the volatility structures are the same. The positivity and significativity of γ_1 in the GJR-GARCH model highlights the presence of leverage effect: negative log-returns are linked with an increase in the volatility level. The possibility of taking into account this phenomenon is the main motivation for the introduction of the GJR-GARCH model.

Finally, one of the most remarkable results is the presence of the risk premium, so that part of the returns of the DAX is determined by the riskiness of the index, for which volatility is a pretty good proxy. The well-established cause-effect relation between high risk and high returns is captured by the GARCH-M model and, in particular, it is highlighted by the significativity and positivity of δ. The E-GARCH model can not account for the leverage effect: indeed, a negative γ_1 would be expected; instead, γ_1 is positive and significant, so it does not realize the negative relation between log-returns and volatility.

5.2. Residual Diagnostic

In this subsection, the residual diagnostic is investigated, by analysing mainly two aspects: namely, in the first subsection, the Gaussianity of the standardized residual is tested (both graphically and analytically with a Jarque Bera test [11]) meanwhile in the second one a test on the residual autocorrelation is done by exploiting the Ljung-Box test.

5.2.1. Gaussianity of the standardized residuals

The results coming from the diagnostic of the standardized residuals give the following results: for what concerns the normality of the standardized residual, a Jarque-Bera test is used after the following standardization procedure:

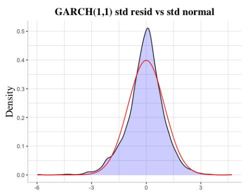
$$v_t = \frac{u_t}{\sigma_t} \tag{10}$$

For all four models, the standardized residuals have a mean very close to 0 and variance very close to 1, but the p-values of the Jarque-Bera test turn out to be very low (less than 2e-16 for all the models) so that the Gaussianity hypothesis has to be rejected. Indeed, the skewness and kurtosis are not so close to the Gaussian values (0 and 3, respectively), even though they are not so far as well. The results are summarized in the following table:

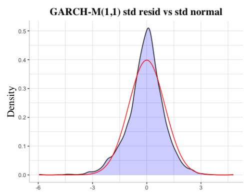
Model and order	Mean	Variance	Skewness	Kurtosis	J-B p-value
GARCH(1,1)	-0.0478	1.0004	-0.3936	4.7390	$<$ 2.2e-16
GARCH-M(1,1)	-0.0579	0.9994	-0.3950	4.7187	$<$ 2.2e-16
E-GARCH(1,1)	0	1.0009	-0.3851	4.5793	$<$ 2.2e-16
GJR-GARCH(1,1)	-0.0108	1.0013	-0.4137	4.6566	$<$ 2.2e-16

Table 11: Gaussianity of the standardized residuals

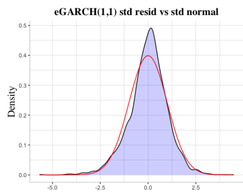
As a further investigation of the Gaussianity of the standardized residuals, a plot of the distributions vs a Standard Gaussian distribution can be done, providing for the four chosen models the following results:



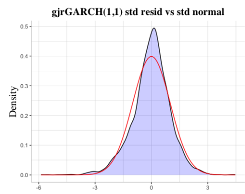
(e) GARCH(1,1)



(f) GARCH-M(1,1)

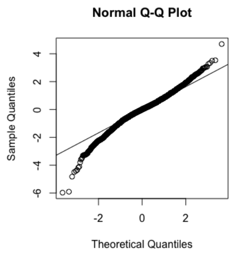


(g) E-GARCH(1,1)

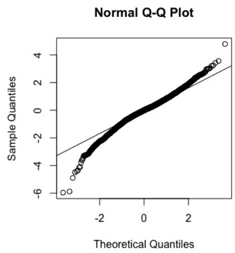


(h) GJR-GARCH(1,1)

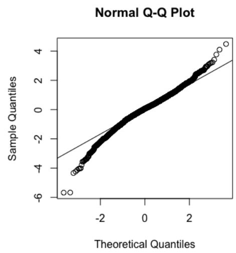
By the previous plot, at least graphically, one can easily observe that the situation has been improved if compared with the distribution of the log-returns vs Standard Normal (see Figure 3). Anyway, there must be a sort of "source of non-normality", and in order to detect one can easily do a QQ-plot:



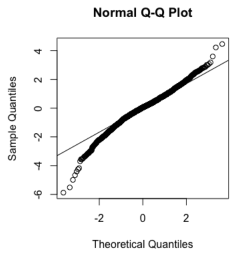
(i) GARCH(1,1)



(j) GARCH-M(1,1)



(k) E-GARCH(1,1)



(l) GJR-GARCH(1,1)

All four Q-Q plots above highlight the presence of a fat left tail, a (less) fat right tail and a slightly leptokurtic distribution (as expected by the value of the kurtosis in Table 9).

5.2.2. Residual Autocorrelation

Moreover, a non-residual-autocorrelation check has been done through a Ljung-Box test [12]. The p-values of the test for each model are:

Model and order	Ljung-Box p-value
GARCH(1,1)	0.9094
GARCH-M(1,1)	0.8687
E-GARCH(1,1)	1
GJR-GARCH(1,1)	0.8618

Table 12: Non-autocorrelation of the standardized residuals

Fortunately, at least this check gives satisfying results. Indeed, in each model, we accept the H_0 hypothesis, which means "no residual autocorrelation". Although the Gaussianity requirements are not satisfied, since the autocorrelation provided good results and the skewness and kurtosis are not so far from the Gaussians ones (see the plots of the distributions below), one can choose to evaluate the forecasting capability of the models as a validation procedure.

Since the diagnostic of the standardized residuals fails, even if better results are obtained concerning the original residuals, it is necessary to use results obtained from robust standard error evaluation. The estimations and their significativities are reported in Tables 7, 10, 8, 9.

5.3. Forecasting

This subsection presents volatility forecasts for the DAX Index during December 2022. In Figures 5, 6, 7 and 8, you can see estimates derived from the selected models (see Subsection 5.1). The dark line in the plots represents the realization of log-returns. In contrast, the three red-shaded areas from darker to lighter indicate the intervals at one, two and three standard deviations from the expected value, respectively. Empirical data stay in the one standard deviation area, and the only outlier is the realized log-return on the 15^{th} of December. The drop affected the whole market due to interest rates increase threats. "*Global stocks dropped and sovereign debt yields climbed yesterday as investors took fright at a wave of interest rate rises from central banks, which threatened more to come in the fight to tame inflation*" writes the Financial Times on the 16^{th} of December talking about the previous trading day. Visually, forecasts are very similar; values rarely exceed one standard deviation from the mean in all estimates, indicating their validity. To better understand which model works better, an error computation is needed. Given a model of the four considered, we define the absolute error of the model as

$$\Delta\sigma \stackrel{\text{def}}{=} \sum_{t \in T} |\sigma_{E_t} - \sigma_{M_t}| \tag{11}$$

where T is the set of trading days in December 2022, which, in the case of the DAX Index, amounts to 21 total dates. σ_{E_t} is the empirical stock volatility at time t and σ_{M_t} is the estimated stock volatility at time t. Recall that, fixing $t \in T$, σ_{E_t} is computed as the absolute value of the realized log-return at time t [11].

Model and order	$\Delta\sigma$
GARCH(1,1)	0.119826
GARCH-M(1,1)	0.119631
E-GARCH(1,1)	0.125437
GJR-GARCH(1,1)	0.127612

Table 13: Absolute error of the four models considered in estimates for December 2022. DAX Index is considered.

Table 13 presents absolute errors for the selected models. Here, GARCH and GARCH-M models present very similar results in accordance with the theory. On the other hand, E-GARCH and GJR-GARCH present slightly bigger errors. If the sole criterion for assessing the performance of a model were the absolute error of predictions, it would be evident that the first two models outperform the others. The reasons for this can be their enhanced model flexibility, the ability to capture volatility clustering (see [2]), and explicit incorporation of the risk-return relationship (see [1]). GARCH models offer greater flexibility by incorporating lagged volatility terms, capturing the autocorrelation and persistence of volatility. This flexibility enables them to capture better the complex volatility patterns observed in financial markets. Additionally, GARCH models excel at capturing volatility clustering, where periods of high volatility tend to cluster together. Moreover, the GARCH-M model explicitly acknowledges the risk-return relationship by incorporating lagged volatility terms in the mean equation, making it more suitable for forecasting log-return volatility and capturing the impact of volatility on expected returns.

Now there is the need to tackle one last issue: what if these errors were just a specific result due to the particular realization of the DAX? Analysis with two other indexes is repeated: NASDAQ Composite Index (IXIC) and Nikkei 225 Index (NI225). The same behaviour can be seen in Table 14 and Table 15, respectively. Reports on the diagnostic and analysis of the results are not presented since they are very similar to the ones of the DAX index.

Model and order	$\Delta\sigma$
GARCH(1,1)	0.167706
GARCH-M(1,1)	0.168780
E-GARCH(1,1)	0.198814
GJR-GARCH(1,1)	0.184415

Table 14: Absolute error of the four models considered in estimates for December 2022. NASDAQ Composite Index is considered.

Model and order	$\Delta\sigma$
GARCH(1,1)	0.135627
GARCH-M(1,1)	0.135712
E-GARCH(1,1)	0.158812
GJR-GARCH(1,1)	0.152337

Table 15: Absolute error of the four models considered in estimates for December 2022. Nikkei 225 Index is considered.

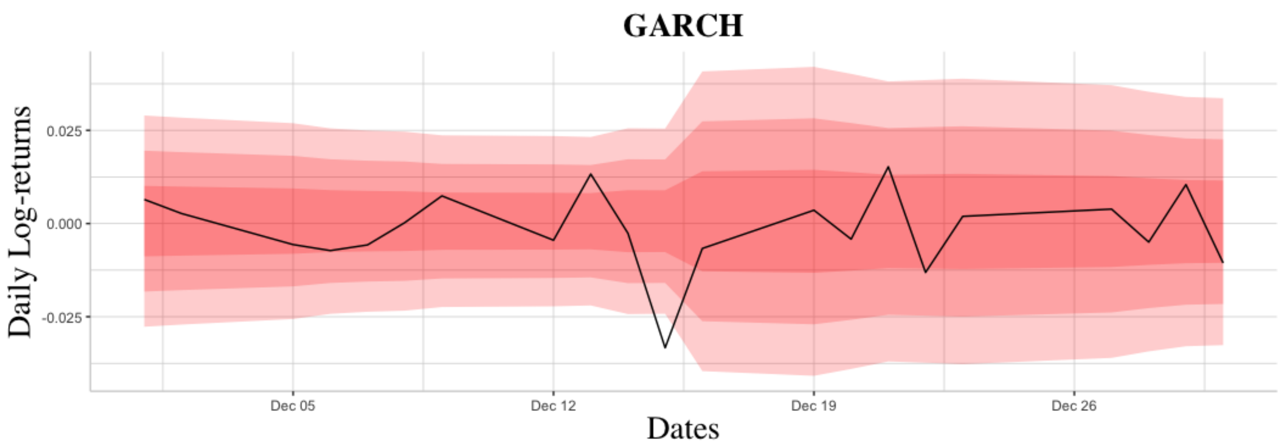


Figure 5: Forecasting stock volatility with GARCH model in December 2022. DAX Index is considered.

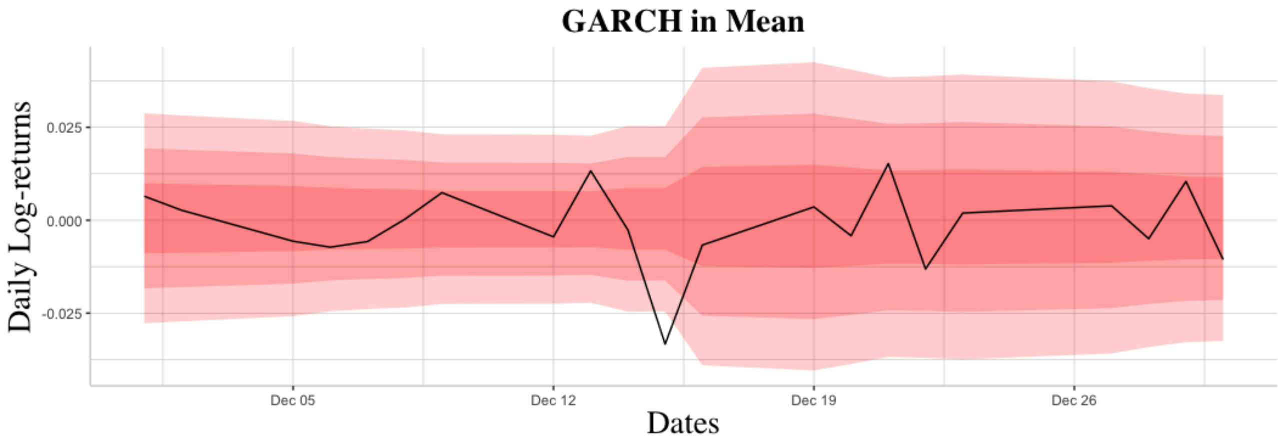


Figure 6: Forecasting stock volatility with GARCH in Mean model in December 2022. DAX Index is considered.

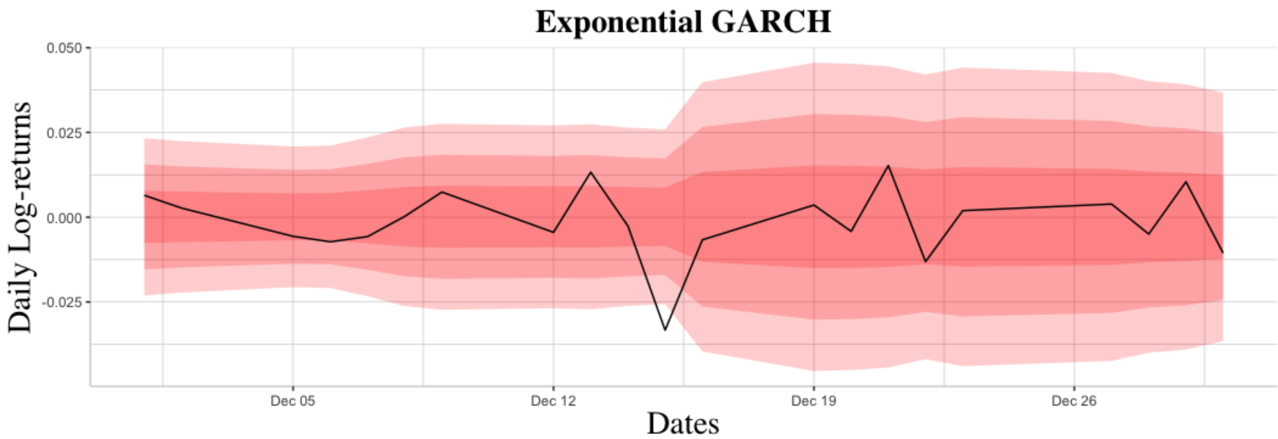


Figure 7: Forecasting stock volatility with Exponential GARCH model in December 2022. DAX Index is considered.

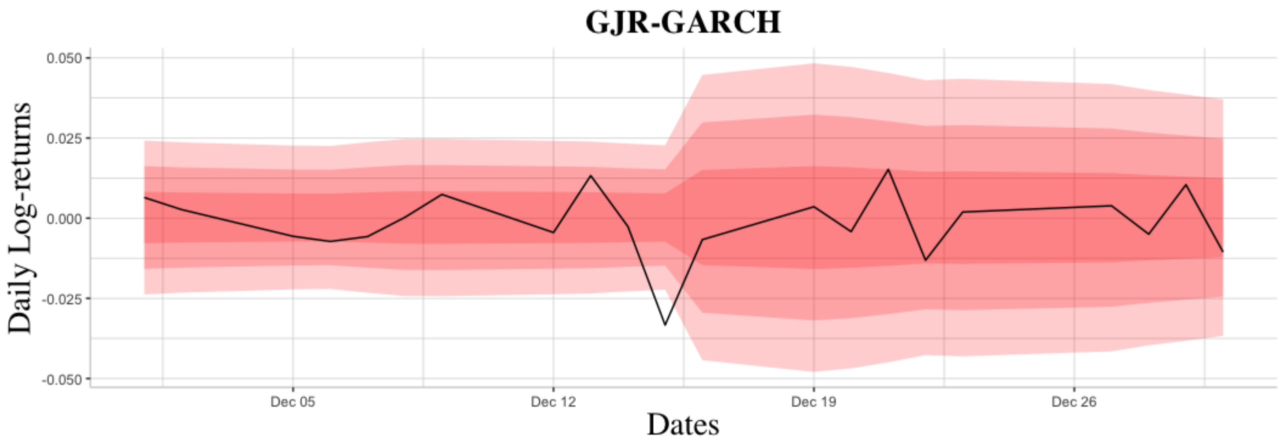


Figure 8: Forecasting stock volatility with Glosten, Jagannathan and Runkle GARCH model in December 2022. DAX Index is considered.

6. Conclusions

This project aimed to seek the best possible model to describe the volatility of the daily log-returns of the DAX index from the 3^{rd} of January 2008 to the 31^{st} December 2022. The aim was to analyze how the features of a solid and stable country's stock market (leverage effect, leptokurtosis, presence/absence of risk premium) are detected by volatility models which are well-known and used in the specific literature. Namely, GARCH model and its extensions have been used to describe the peculiar characteristics of the volatility behaviour mentioned above.

The main results found are the presence of the leverage effect and a risk premium component, which was expected. Moreover, the forecasting results highlight the fitted models' reliability in predicting future volatilities. Apart from very few outliers, the predicted volatilities describe the daily log-returns oscillations quite well. The only unsatisfactory result concerns the residuals' diagnostic, which is not such a big issue considering the goodness of the forecasting.

A further exciting extension of this project can be carried out by including other external regressors in the GARCH models. For instance, some macroeconomic variables can be considered in the analysis (inflation, dividend yield, default return spread, etc.) as is done in [11].

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